



Bansilal Ramnath Agarwal Charitable Trust's Vishwakarma
Institute of Information Technology

Department of
Artificial Intelligence and Data Science

Student Name : Sathe Onkar Abasaheb

Class: SY

Division:B

Roll No : 272078

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Subject Name & Code: AD22232

Assignment No: 5

Title of Assignment: Identify Analysis level classes and Design Class Diagrams.

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Date of Submission: 03 /10/2025

About class diagram :-

In software engineering a class diagram in the unified modelling language (UML) is a type of structural static structure diagram that describes the structures of the system by showing operations (methods) and the relationships amongst objects.

2. Components of class diagram.

Upper Section - contains the name of the class.
This section is always req^d whether you are talking about the class or an object.

Middle Section - Contains attributes of the class.
Use this section to describe the quantities of the class.

Bottom Section - Include class operations (methods) displayed in the format each operation takes its own line. The operations describe how a class interacts with the data.

Relationship In class diagram.

1) Inheritance - Inheritance is also called as generalization and is used to describe the relationship between parent and child classes.

2) Implementation - It's mainly used to specify the relationship between the interfaces and implementation class.

3) Composition Relationship - The Relationship between the whole and the part but the whole and part can't be separated.

4) Aggregation Relationships -
The Relationship between the whole and part and the whole and part can be separated.

5) Association Relationships.
Indicates that a property of the class holds a reference to an instance of another class.

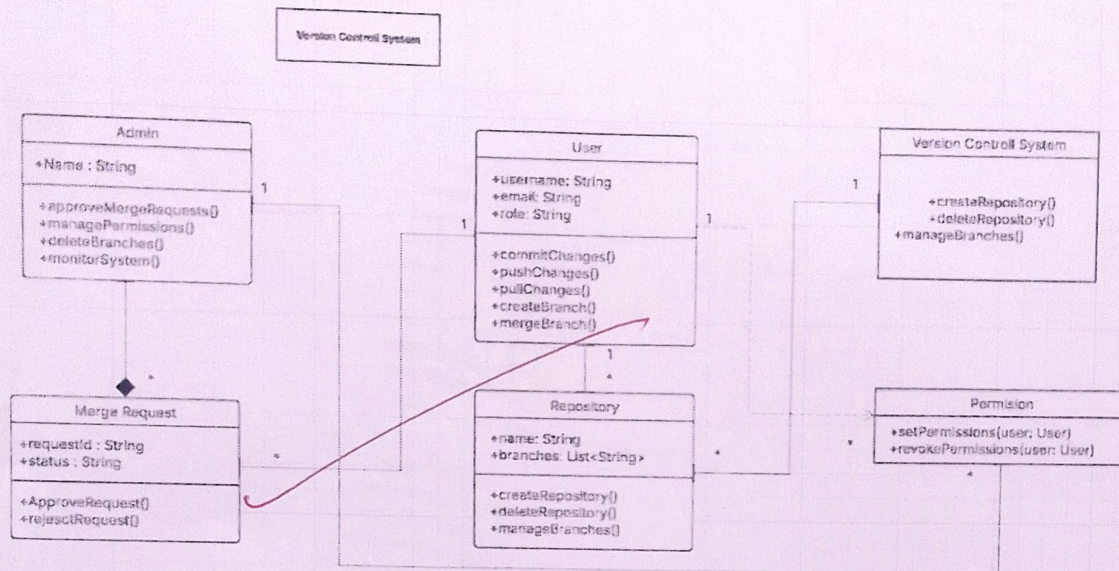
6) Dependencies.

- It's reflected in method of a class that we use another class's object as a parameter.

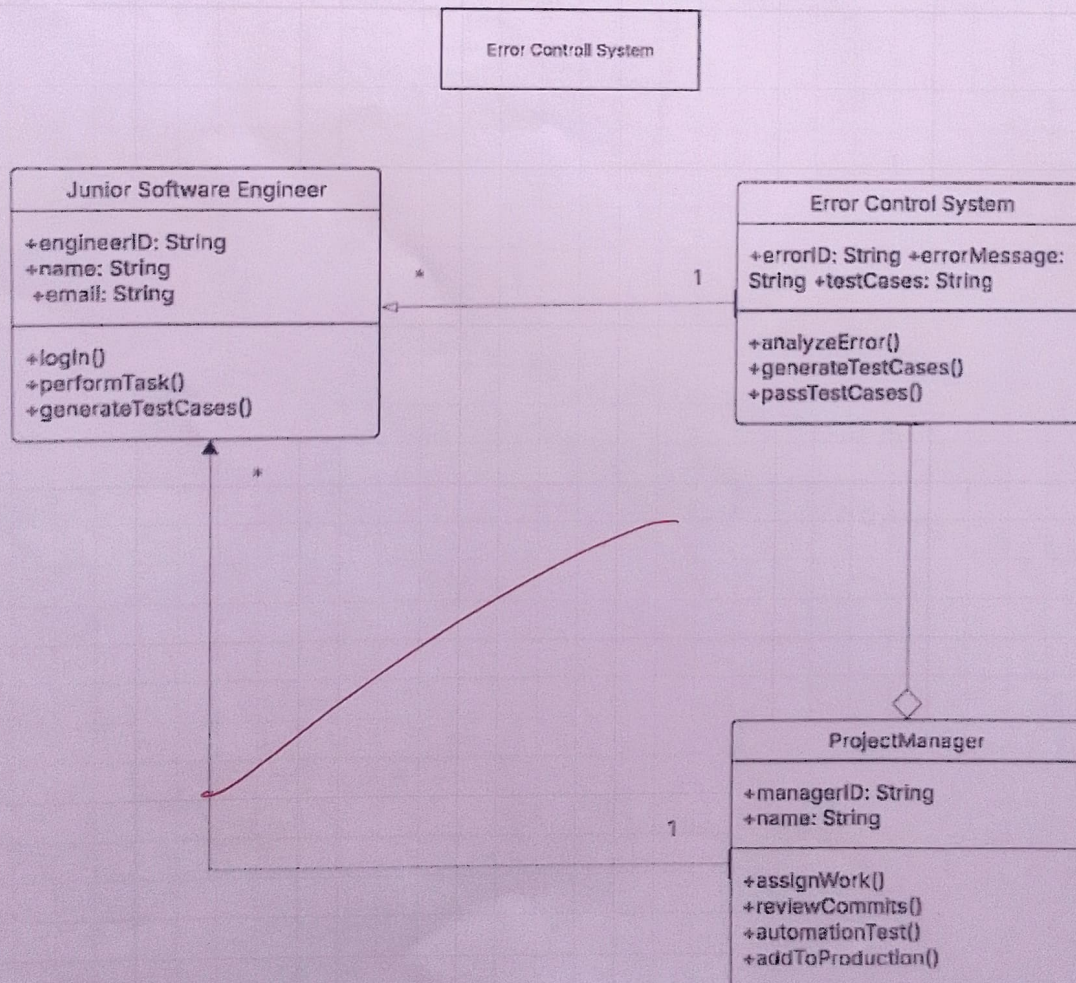
Conclusion

Here we learned to draw class diagrams along with OOP concepts.

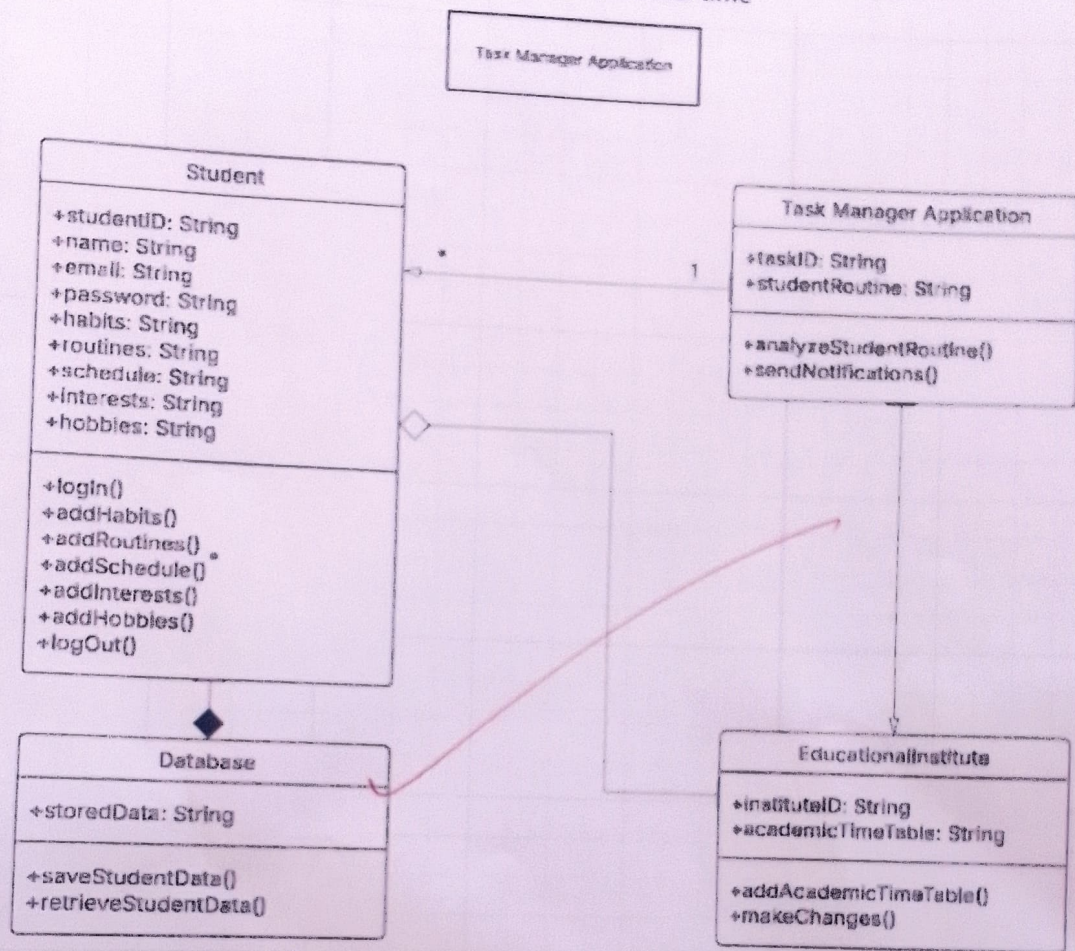
CPS 1: Freelancer trying to manage, collaborate and track the changes in code



CPS 2: Junior Software Developer engineer trying to fix errors in code and make it work correctly.



CPS3: Balance student academic responsibilities with personal time



Concluⁿ - here we have learned to create a class diagram with the OOP concepts like class, methods, inheritance, access specifier etc.

Pranav ali
11/3/2025